TP53
BRCA1
EGFR
MYC
STAT3
CDK1
. . .

Gene Annotation & Downstream Analysis of Differentially Expressed Genes

From Differential Expression to Biological Insights

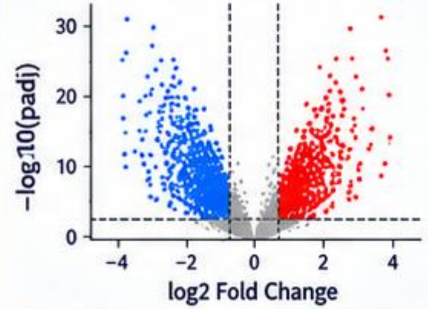
RNA-seq Analysis Workshop | Bioinformatics & Computational Biology

Gene Identification → Functional Annotation → Pathway Analysis → Biological Interpretation

Genes in Context for a Healthier Tomorrow.

- Function
- Pathways
- Biomarkers
- Precision Medicine

1 Differentially Expressed Genes



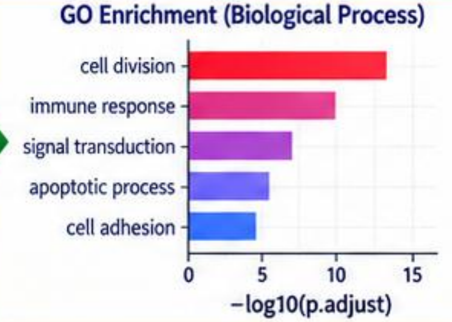
Identify significantly upregulated and downregulated genes

2 Gene Annotation

GENCODE
Ensembl
NCBI
UniProt

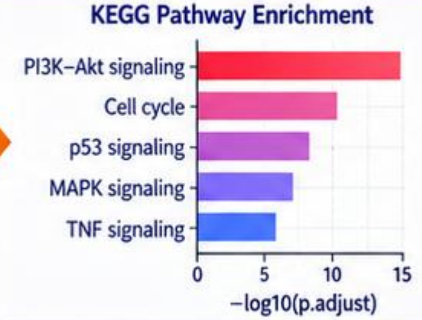
Annotate DEGs with functional information (gene name, function, chromosome, biotype, etc.)

3 Functional Enrichment Analysis



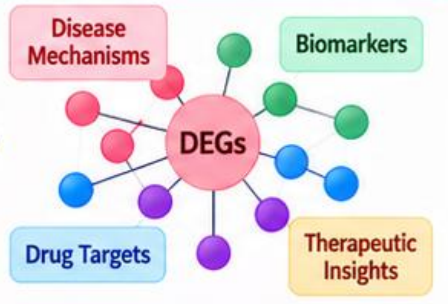
Identify overrepresented biological processes, molecular functions and cellular components

4 Pathway Analysis



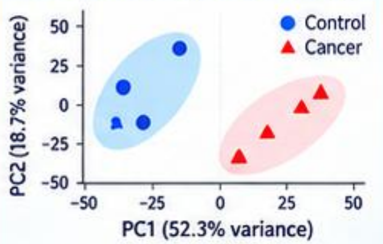
Identify significantly enriched pathways and molecular networks

5 Biological Interpretation

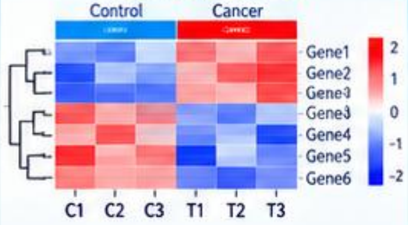


Integrate results to understand biological meaning and identify key candidate genes

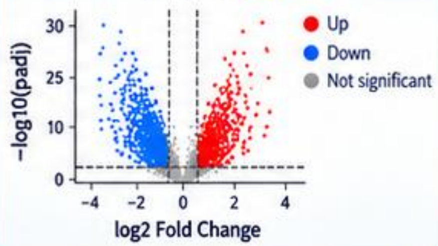
PCA Plot (Sample Clustering)



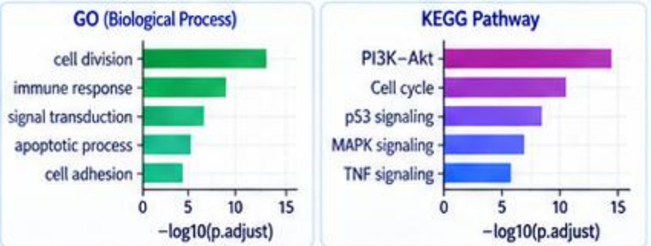
Heatmap of DEGs



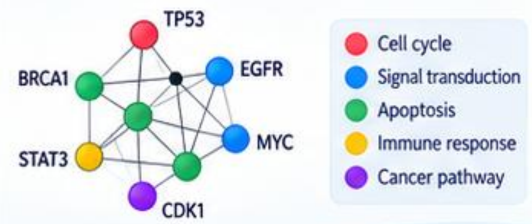
Volcano Plot



Functional Enrichment



Protein–Protein Interaction Network



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NIELIT
— Empowering Through ICT —

By
Manabendra Nath
Senior Bioinformatician
NIELIT, Guwahati

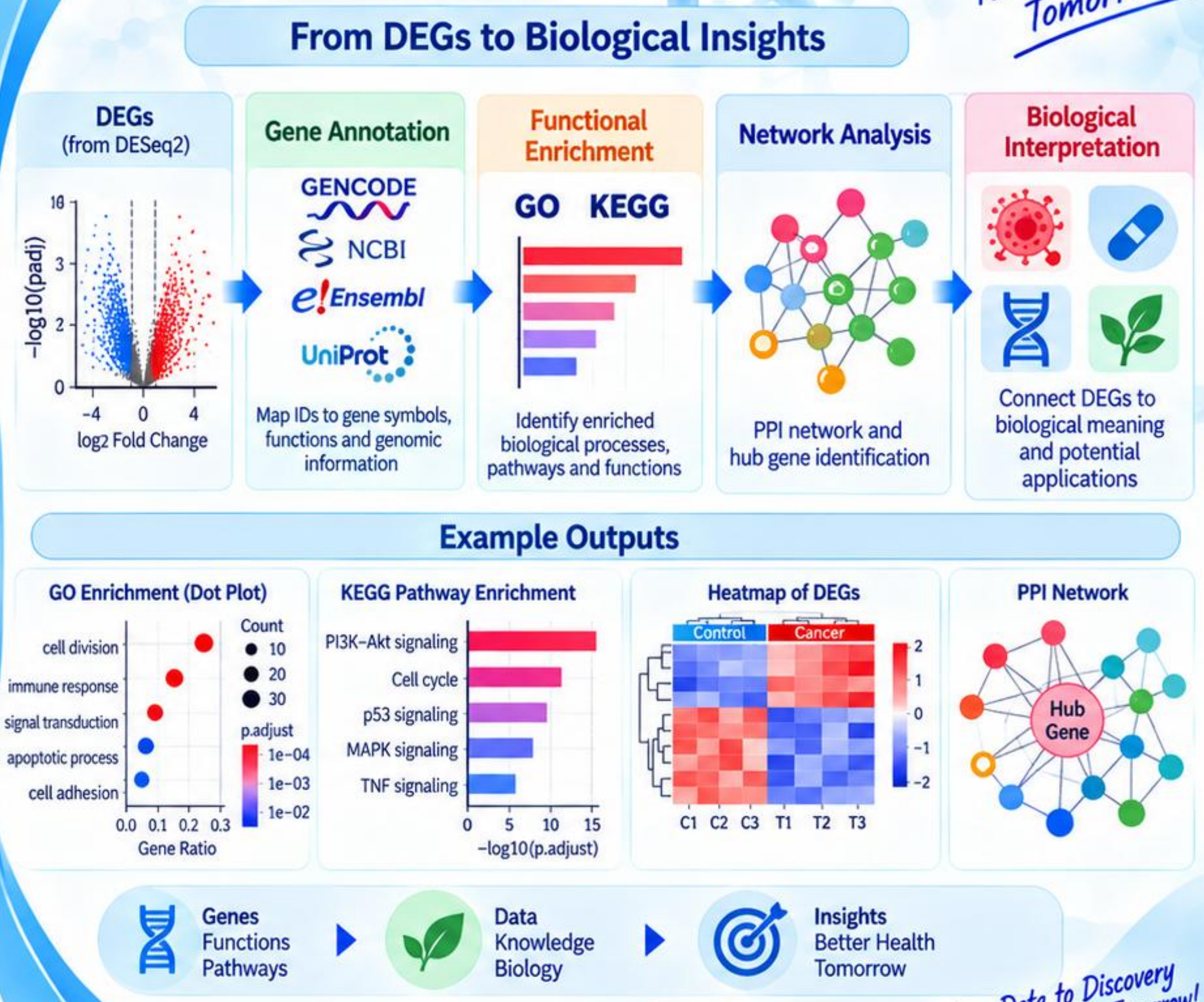
Data to Discovery for a Better Tomorrow

LEARNING OBJECTIVES

By the end of this session, participants will be able to:

1. **Understand** the concept and importance of gene annotation after DEG analysis.
2. **Identify** different gene identifiers and map them to gene symbols and functional information.
3. **Perform** functional annotation of differentially expressed genes using Gene Ontology (GO).
4. **Analyze** enriched biological pathways using KEGG, Reactome, and other pathway databases.
5. **Interpret** enrichment results to connect DEGs with biological processes and molecular functions.
6. **Visualize** functional and pathway enrichment results using publication-quality plots.
7. **Explore** protein–protein interaction (PPI) networks and identify potential hub genes.
8. **Integrate** DEG, functional, and pathway information to derive biological insights.

Genes in Context for a Healthier Tomorrow



From Data to Discovery for a Healthier Tomorrow!

3 What is Gene Annotation?

Gene annotation = assigning biological information to a gene

It provides information about:

- **Gene identity** (e.g., Ensembl ID, Entrez ID)
- **ABC Gene symbol / name** (e.g., TP53)
- **Gene function** (what the gene does)
- **Protein product** (protein sequence, structure)
- **Cellular location** (e.g., nucleus, cytoplasm)
- **Biological processes** (e.g., cell cycle, apoptosis)
- **Molecular functions** (e.g., DNA binding, kinase activity)
- **Pathways** (e.g., KEGG, Reactome)
- **Protein domains** (e.g., Pfam, InterPro)
- **Disease associations** (e.g., cancer, genetic disorders)

A Gene
A Function
A Bigger Story!

Simple concept

Gene ID (ENSG, Entrez, etc.)	Gene identity (symbol/name)	Biological function	Pathway & disease link
ENSG00000141510	TP53	Tumor suppressor	Cell cycle, p53 pathway

4 Why Do We Need Gene Annotation?

After DEG analysis, we may obtain:

From IDs to Insights!

List of DEGs (Gene IDs)

ENSG00000141510
ENSG00000146648
ENSG00000171862
ENSG00000155657
ENSG00000198727
ENSG00000204287
ENSG00000130203
ENSG00000157764
...

? These IDs alone do not tell us much about their biological roles.

Annotation converts

Annotated Information

Gene ID	Gene Symbol	Function / Description
ENSG00000141510	TP53	Tumor suppressor protein
ENSG00000146648	BRCA1	DNA repair protein
ENSG00000171862	EGFR	Receptor tyrosine kinase
ENSG00000155657	MYC	Transcription factor
...	...	...

DEG analysis tells us **which genes changed,**
while annotation tells us **what those genes are and what they do.**

DEGs (Which genes?) + **Annotation** (What are they?) = **Biological Insights** (Why it matters?)

Annotate Today
Understand Tomorrow

5 Basis of Gene Annotation

Gene annotation is based on reference databases and biological evidence.

Genome sequence Reference genome (e.g., GRCh38)	Provides the genomic coordinates and sequence information.
Gene Genomic locus (e.g., gene model)	Defines the gene structure (exons, introns, boundaries).
Transcript mRNA / isoform (e.g., ENST ID)	Alternative splicing generates multiple transcripts.
Protein Protein product (e.g., UniProt ID)	Translated protein sequence and structure.
Known biological information From databases, literature and experiments	Function, domains, pathways, interactions, disease associations, etc.
Functional annotation Assign biological meaning to the gene	Integrate evidence to provide comprehensive annotation.

Annotation is supported by:

- **Experimental evidence** Function validated in laboratory experiments (e.g., enzyme activity, phenotypes).
- **Sequence similarity** Homology to known genes/proteins (e.g., BLAST).
- **Orthology** Conserved genes across species (infers function).
- **Protein domains** Conserved domains and motifs (e.g., Pfam, InterPro).
- **Curated literature** Manually curated information (e.g., UniProt, NCBI, GeneCards).
- **Computational prediction** In silico methods (e.g., gene structure prediction, function prediction, GO inference).

Multiple sources of evidence build reliable annotation!

From Sequence to Function
Reference Databases + Biological Evidence = Functional Annotation

6 Different Types of Annotation

Different annotation types provide complementary information about a gene and its products.

Annotation type	Information obtained	Example	Database / Tool
Gene annotation	Gene identity (gene symbol, name, chromosome, location)	TP53 chr17:7,566,964-7,590,856	GENCODE, NCBI
Transcript annotation	Transcript/isoform information (exon-intron structure, isoforms)	ENST00000269305	GENCODE, Ensembl
Protein annotation	Protein function (protein sequence, structure, name)	P04637 (tumor protein p53)	UniProt
GO annotation	Biological function (BP, MF, CC)	GO:0006915 (apoptotic process)	Gene Ontology
Pathway annotation	Biological pathways	hsa04115 (p53 signaling pathway)	KEGG, Reactome
Domain annotation	Protein domains and family information	PF00870 (p53 DNA-binding domain)	Pfam, InterPro
Disease annotation	Disease associations (e.g., cancers, genetic disorders)	Breast cancer, Li-Fraumeni syndrome	DisGeNET, GeneCards

Integrating multiple annotation types provides a comprehensive view of gene function and its role in biology.

More Knowledge Deeper Insights!

8 Major Annotation Databases

Multiple Databases
A Complete View

Different databases provide complementary information about genes and proteins

Ensembl
- Gene and transcript information
- Genomic coordinates
- Gene identifiers (ENSG, ENST, etc.)
- Comparative genomics

Genome annotation and gene models

NCBI
- Gene information
- Gene IDs
- RefSeq sequences
- Genomic information
- Link to other databases

Comprehensive genetic information

UniProt
- Protein information
- Protein function
- Protein domains
- Subcellular localization
- Post-translational modifications
- Interactions
- Disease relevance

Protein functional annotation

GO Gene Ontology
- Biological processes (BP)
- Molecular functions (MF)
- Cellular components (CC)
- Controlled vocabulary
- Hierarchical structure

Functional description of genes

KEGG / Reactome
- Biological pathways
- Molecular interactions
- Metabolic pathways
- Signaling pathways
- Disease pathways

Pathway and network information

From Gene IDs to Biological Meaning

9 UniProt Annotation

Proteins
Give Life to Genes

What can UniProt tell us?

A gene/protein can be associated with:

- Protein name
- Function
- Gene name
- Subcellular location
- Protein domains
- Post-translational modifications
- Interactions
- Disease relevance
- Sequence information
- Cross-references (to other databases)

Example: TP53 in UniProt

Gene	TP53
Protein	Tumor protein p53
Function	DNA damage response
Biological role	Cell-cycle regulation Apoptosis

UniProt provides detailed protein-level information that helps understand gene function.

10 Gene Ontology (GO)

A Common Language
for Gene Function

Gene Ontology (GO) provides standardized descriptions of gene function.

GO
Gene Ontology
Unifying Biology

- Controlled vocabulary
- Hierarchical structure
- Used across species
- Describes gene products

Three major categories

Biological Process (BP)
What biological process does the gene participate in?

Examples:
- Cell cycle
- Apoptosis
- DNA repair
- Immune response

Molecular Function (MF)
What does the gene/protein do?

Examples:
- DNA binding
- ATP binding
- Protein kinase activity
- Transcription factor activity

Cellular Component (CC)
Where does the gene/protein act?

Examples:
- Nucleus
- Mitochondrion
- Plasma membrane
- Cytoplasm

11 GO Annotation vs GO Enrichment

These are different concepts.

GO Annotation
Assigns GO terms to a gene.

Gene → GO terms

Example: TP53
- DNA binding (MF)
- Apoptotic process (BP)
- Nucleus (CC)

Describes the function of individual genes.

GO Enrichment
Identifies overrepresented GO terms in a list of genes (e.g., DEGs).

DEG list → GO analysis → Enriched GO terms

Example: DEGs
- Cell cycle ↑↑
- DNA replication ↑
- DNA repair ↑

Finds common biological themes in a gene set.

12 Why Perform Functional Enrichment?

From Gene Lists to Big Insights

A DEG list may contain hundreds or thousands of genes.

↓

Looking at every gene individually is difficult.

↓

Enrichment analysis asks:
"Are particular biological functions or pathways represented more than expected in my DEG list?"

↓

This allows us to move from:

Individual genes
→ Biological processes
→ Biological pathways

Data to Biology

Gene Annotation of Differentially Expressed Genes

From Differential Expression to Biological Insights

DEGs → Annotation → Pathways → Insights

RNA-seq Analysis Workshop
Bioinformatics & Computational Biology

2 Overall Workflow

DEG Dataset
↓
Filter Significant DEGs
↓
Obtain Protein Sequences
↓
BLAST against UniProt
↓
Download UniProt Annotation
↓
Merge and Interpret Results

3 Step 1: Inspect the DEG Dataset

Check the identifiers in your DEG file.

GeneID	Gene Symbol	log2FC	pvalue	padj
ENSG00000141510	TP53	2.31	0.0001	0.001
ENSG00000146648	EGFR	1.74	0.0005	0.006
ENSG00000171862	PTEN	-1.82	0.0002	0.003

Important: Determine which type of identifiers you have:

- Ensembl (ENSG...)
- Entrez (e.g., 7157)
- Gene symbol (TP53)
- RefSeq (e.g., NM_000546)
- UniProt (e.g., P04637)

4 Why Do We Need Annotation?

DEG analysis gives us a list of gene IDs, but they do not tell us much about their biological roles.

Annotation converts identifiers to biological information

ENSG00000141510
ENSG00000146648
ENSG00000171862
...

→

- Gene name
- Function
- Protein product
- Pathways
- Cellular location
- Disease association

DEG analysis tells us which genes changed, while annotation tells us what those genes are and what they do.

5 Major Annotation Databases

Ensembl
Gene & transcript info
- Genomic coordinates
- Gene identifiers

NCBI
Gene information
- Gene IDs
- RefSeq
- Genomic information

UniProt
Protein information
- Protein function
- Protein domains
- Subcellular localization

GO Gene Ontology
- Biological functions
- Molecular functions
- Cellular components

KEGG
Biological pathways

Reactome
Biological pathways

6 Step 2: Filter Significant DEGs

Significance criteria:

- $padj < 0.05$
- $|log2FoldChange| \geq 1$ (optional)

Separate the genes into:

UP-regulated DEGs | **DOWN-regulated DEGs**

Why?
To focus downstream analysis on biologically significant genes.

7 Step 3: Identify the Corresponding Protein

A gene is not directly submitted to protein BLAST.
We need to map:

Gene (ENSG00000141510)
↓
Transcript (ENST...)
↓
Protein (ENSP...)
↓
Protein sequence (FASTA)

Example:
ENSG00000141510
↓
TP53
↓
TP53 protein
↓
Protein FASTA

8 Step 4: Obtain Protein FASTA

Possible sources:

- Ensembl
- NCBI
- UniProt
- GENCODE

Output:
DEG_proteins.fasta

9 What is FASTA?

FASTA is a standard format for biological sequences.

```
>Protein_ID
MEEPQSDPSVEPPLSQETFSDLWKLLPENNV...
```

Header (>Protein_ID)

Sequence (amino acid sequence)

10 Step 5: Why BLAST?

BLAST = Basic Local Alignment Search Tool
Compares your protein sequence against known sequences.

Your protein (FASTA) → **BLASTP** → **Reference protein database (UniProt)** → Hit 1, Hit 2, Hit 3, Hit 4

Purpose: To identify similar known proteins.

11 BLASTP vs BLASTN

BLAST type	Query	Used for
BLASTN	DNA/RNA	Nucleotide comparison
BLASTP	Protein	Protein comparison
BLASTX	DNA → protein	Translation-based search
TBLASTN	Protein → DNA	Protein against translated DNA

For our workflow:
Protein FASTA → BLASTP

12 Step 6: BLAST Against UniProt

Use the UniProt BLAST service.

Input
DEG_proteins.fasta

→

Search against UniProt protein database

UniProt

Output provides:

- UniProt accession
- Protein name
- Organism
- Sequence identity
- Query coverage
- E-value
- Alignment

13 Understanding BLAST Results

Important parameters:

Percentage Identity	99% → very high similarity 70% → substantial similarity 30% → careful interpretation
Query Coverage	100% → entire sequence 50% → half of sequence
E-value	Lower E-value = stronger evidence (e.g., 0, 1e-50)

14 How to Select the Best BLAST Hit

Consider multiple criteria:

1. Percentage identity
2. Query coverage
3. E-value
4. Protein length
5. Organism (e.g., *Homo sapiens*)
6. Protein name (relevant function)
7. Reviewed status
8. Conserved domains

Ideal example:
Identity = 98%
Coverage = 100%
E-value = 0
Organism = Homo sapiens
Reviewed = Yes

15 Reviewed vs Unreviewed UniProt

Swiss-Prot (Reviewed)
- Manually curated
- Higher-quality annotation
- Literature-supported information

TrEMBL (Unreviewed)
- Computationally annotated
- May not have extensive manual curation

Recommendation:
Prefer a reviewed UniProtKB/Swiss-Prot entry when available.

16 Step 7: Record the Best BLAST Hit

Gene	UniProt	Identity	Coverage	E-value	Reviewed
TP53	P04637	99.2%	100%	0	Yes
EGFR	P00533	100%	100%	0	Yes
PTEN	P60484	98.5%	99%	1e-50	Yes
...	...	...	...	...	...

Save as: BLAST_results.tsv

17 Step 8: Obtain UniProt Annotation

Use the UniProt accession obtained from BLAST.

P04637
P00533
P60484
...

→

Search in UniProt and download TSV file

UniProt

Output: UniProt_annotation.tsv

18 What Information Can Be Downloaded?

Basic annotation
- UniProt accession
- Protein name
- Gene name
- Organism
- Protein length

Functional annotation
- Function
- Catalytic activity
- Subcellular location
- Pathway

Cross references
- Ensembl
- NCBI Gene
- RefSeq
- KEGG
- PDB

19 Gene Ontology Information

Biological Process (BP)
- DNA repair
- Cell cycle
- Signal transduction
- ...

Molecular Function (MF)
- DNA binding
- ATP binding
- Kinase activity
- ...

Cellular Component (CC)
- Nucleus
- Cytoplasm
- Plasma membrane
- ...

20 Protein Domain Annotation

InterPro
Protein families and domains

Pfam
Protein domains

PROSITE
Functional sequence patterns/sites

Helps answer:
What structural or functional regions does this protein contain?

21 Step 9: Download UniProt TSV

Recommended fields:

- Entry
- Entry Name
- Reviewed
- Protein names
- Gene Names
- Organism
- Length
- Function
- Subcellular location
- Pathway
- GO Biological Process
- GO Molecular Function
- GO Cellular Component
- InterPro
- Pfam
- Ensembl
- RefSeq

TSV

22 Step 10: Merge the Two Files

File 1: BLAST_results.tsv
- GeneID
- Gene_symbol
- UniProt_Accession
- Identity
- Coverage
- E-value
- ...

+

File 2: UniProt_annotation.tsv
- Entry
- Protein name
- GO
- InterPro
- KEGG
- Function
- ...

23 Merge Key

Common identifier:

BLAST_results
UniProt_Accession

→

UniProt annotation
Entry

↓

Use this as the merge key.

UniProt_Accession = Entry

24 Final Annotated DEG Table

Gene	log2FC	padj	UniProt	Protein	GO-BP	GO-MF	GO-CC	KEGG
TP53	2.31	0.001	P04637	p53	DNA repair	DNA binding	Nucleus	hsa04115
EGFR	1.74	0.006	P00533	EGFR	signaling	kinase activity	Membrane	hsa04012
PTEN	-1.82	0.003	P60484	PTEN	cell cycle	ATP binding	Cytoplasm	hsa04110
...	...	...	...	...	...	...	...	...

22 Step 10: Merge the Two Files

Recommendad fields:

- Entry
- Entry Name
- Reviewed
- Protein names
- Gene Names
- Organism
- Length
- Function
- Subcellular location
- Pathway
- GO Biological Process
- GO Melecular Function
- GO Cellular Component
- InterPro
- Pfam
- Ensembl
- RefSeq

TSV

Output: UniProt_annotation.tsv

23 Merge Key

Common identifier:

BLAST_results
UniProt_Accession

→

UniProt annotation
Entry

↓

Use this as the merge key.

UniProt_Accession = Entry

24 Final Annotated DEG Table

Gene	log2FC	padj	UniProt	Protein	GO-BP	GO-MF	GO-CC	KEGG
TP53	2.31	0.001	P04637	p53	DNA repair	DNA binding	Nucleus	hsa04115
EGFR	1.74	0.006	P00533	EGFR	signaling	kinase activity	Membrane	hsa04012
PTEN	-1.82	0.003	P60484	PTEN	cell cycle	ATP binding	Cytoplasm	hsa04110
...	...	...	...	...	...	...	...	...

This becomes your **Master DEG Annotation Table.**

25 Why Keep DEG Statistics?

Do not remove:

- log2FoldChange
- p-value
- adjusted p-value
- baseMean
- regulation

Because later we can connect:
Differential expression
+
Protein annotation
+
Functional annotation
↓
Biological interpretation

26 From Data to Biological Insights

Differential Expression (DEGs)
↓
Gene Annotation (UniProt)
↓
GO / KEGG / Pathways
↓
Biological Interpretation

Genes Tell a Story When We Decode Them!

1 Ensembl: Basic Concept

e! Ensembl

Genome Information for a Better Tomorrow

What is Ensembl?

Ensembl is a **genome annotation resource** that provides information about:

- **Genes** – Gene models and sequences
- **Transcripts** – Transcript models and isoforms
- **Proteins** – Protein sequences and domains
- **Genomic locations** – Chromosomal coordinates
- **Gene IDs** – e.g., ENSG... (Ensembl Gene ID)
- **Transcript IDs** – e.g., ENST... (Ensembl Transcript ID)
- **Protein IDs** – e.g., ENSP... (Ensembl Protein ID)
- **Functional annotations** – GO, pathways, domains, variants, etc.

Comprehensive Genome Annotation for All Organisms!

Main purpose

To establish the relationship:

Genome → Gene → Transcript → Protein → Function

Example: TP53

Ensembl Gene ID	Gene Symbol	Transcript ID (Example)	Protein ID (Example)	TP53 protein
ENSG00000141510	TP53	ENST00000269305	ENSP00000269305	Tumor protein p53

Ensembl links genomic information to biological function.
From sequence to function — Ensembl provides the complete picture!

2 Ensembl: Methodology

Explore Genomes
Understand Biology
Enable Discovery

Workflow

1. **DEG dataset** – List of differentially expressed genes (e.g., ENSG IDs)
2. **Identify Ensembl Gene IDs** – Check if your DEGs contain Ensembl IDs (ENSG...)
3. **Ensembl database** – Access Ensembl (website, BioMart, or API)
4. **Retrieve gene information** – Gene symbol, description, biotype, etc.
5. **Retrieve transcript information** – Transcript ID, transcript biotype, exon structure
6. **Retrieve protein ID** – Protein ID, protein sequence, domains
7. **Retrieve functional annotations** – GO, pathways, cross-references, etc.
8. **Merge with DEG table** – Combine Ensembl annotation with your DEG dataset

Output: Information Retrieved from Ensembl

Ensembl Gene ID	(e.g., ENSG00000141510)
Gene symbol	(e.g., TP53)
Transcript ID	(e.g., ENST00000269305)
Protein ID	(e.g., ENSP00000269305)
Gene description	(e.g., tumor protein p53)
Chromosomal location	(e.g., 17p13.1)
GO annotations	Biological Process (BP) Molecular Function (MF) Cellular Component (CC)
Pathway information	(e.g., KEGG, Reactome)
Cross-references	(e.g., UniProt, RefSeq, NCBI Gene, HGNC, PDB, etc.)

From Genome to Function
Ensembl provides the genomic, transcriptomic, proteomic and functional context in a single integrated resource.

e! Ensembl

Open Data for a Healthier World

1 Gene Ontology (GO) Basic Concept

What is Gene Ontology?

Gene Ontology (GO) is a standardized system for describing the biological functions of genes and proteins.
It provides a common vocabulary that can be applied across different organisms.

Gene Ontology
Unifying Biology

GO answers three questions:

1. What biological process does the gene participate in? → **Biological Process (BP)**
2. What molecular activity does the gene/protein perform? → **Molecular Function (MF)**
3. Where does the gene/protein function? → **Cellular Component (CC)**

A common language for gene function across species

2 Three Ontologies of GO

1 Biological Process (BP)
Describes biological processes or pathways in which a gene participates.

Examples:
- Cell cycle
- DNA repair
- Apoptosis
- Cell proliferation
- Signal transduction
- Immune response

2 Molecular Function (MF)
Describes the molecular activity of a gene product.

Examples:
- ATP binding
- DNA binding
- Protein kinase activity
- Transcription factor activity
- Catalytic activity

3 Cellular Component (CC)
Describes where the gene product is located or functions.

Examples:
- Nucleus
- Cytoplasm
- Plasma membrane
- Mitochondrion
- Ribosome

3 GO Hierarchical Structure

GO terms are organized as a hierarchy.
A more specific GO term is connected to broader parent terms.

- Biological Process (GO:0008150)
  - Cellular process (GO:0009987)
    - Cell cycle (GO:0007049)
    - Cell division (GO:0051301)
  - DNA metabolic process (GO:0006259)
    - DNA repair (GO:0006281)
    - DNA replication (GO:0006260)

GO is not simply a flat list of functional labels. It is an **ontology**, meaning that **relationships** exist **between** terms.

4 Why Perform GO Analysis?

A DEG analysis may give:

500 significant genes

But a list of 500 genes is difficult to interpret biologically.

GO analysis converts:

Large DEG list → Functional categories → Biological interpretation

Example:
500 DEGs
↓
DNA repair
Cell cycle
Apoptosis
Cell proliferation
↓
Potential biological mechanisms

GO helps us move from a list of genes to an understanding of biological processes.

5 GO Annotation vs GO Enrichment

GO Annotation
Determines which GO terms are associated with individual genes.

Example: TP53 → DNA damage response (BP); DNA binding (MF); Nucleus (CC)

GO Enrichment
Determines whether particular GO terms occur more frequently in your DEG list than expected by chance.

Example: 500 DEGs → DNA repair genes → Significantly enriched FDR = 0.001

Annotation ≠ Enrichment

6 GO Enrichment: Basic Principle

Suppose your experiment produces:

Total genes detected = 20,000
Significant DEGs = 500

Suppose:

Genes associated with DNA repair = 300
DNA repair genes among DEGs = 50

We ask:

Is 50 DNA-repair genes unusually high compared with what would be expected by chance?

If statistically significant:
↓
DNA repair is enriched.

7 GO Enrichment: Input

Typical input:

DEG list (e.g.)
- TP53
- BRCA1
- BRCA2
- RAD51
- ATM
- CHEK1
- ...

Required information:
- Gene IDs
- Organism
- Background/universe gene set

Examples of IDs:
- Ensembl
- Entrez
- Gene Symbol
- UniProt

Use appropriate gene identifiers and organism information.

8 Background Gene Set

What is the background?
The background/universe is the set of genes that could have been selected as DEGs.

For RNA-seq, a good background is usually the set of genes that were successfully tested/detected in the experiment.

Genome 20,000 genes
↓
Genes detected/tested 15,000 genes
↓
DEGs 500 genes

The enrichment test compares:
500 DEGs vs 15,000 tested genes
(rather than necessarily the entire genome).

9 GO Enrichment Workflow

DEG Dataset
↓
Filter significant
↓
Gene IDs
↓
ID conversion
↓
Define background
↓
GO database (e.g., GO, org.Hs.eg.db)
↓
Enrichment analysis
↓
BP | MF | CC
↓
Multiple testing
↓
Significant GO terms
↓
Visualization

10 Statistical Principle of GO Enrichment

A commonly used approach is the hypergeometric test or equivalent over-representation test.
The basic question is:

Given the number of genes in the DEG list and the number associated with a GO term, how likely is this overlap to occur by chance?

Hypergeometric probability:

$$P(X=k) = \frac{\binom{K}{k}\binom{N-K}{n-k}}{\binom{N}{n}}$$

Where:
- N = total genes in background
- K = genes associated with the GO term
- n = number of DEGs
- k = DEGs associated with that GO term

11 Multiple Testing Correction

GO analysis tests hundreds or thousands of GO terms. Therefore, many terms can appear significant simply by chance.

Solution:

Apply multiple-testing correction.

A common method is:

Benjamini–Hochberg FDR (False Discovery Rate)

Instead of relying only on:

$p < 0.05$

we commonly examine:

FDR / adjusted p-value < 0.05

This controls the proportion of false positives among the significant results.

12 Important GO Result Columns

Column	Meaning
GO ID	Unique GO identifier (e.g., GO:0006281)
Description	GO term name
GeneRatio	DEGs associated with term / total DEGs
BgRatio	Background genes associated with term / total background
pvalue	Statistical significance
p.adjust	Multiple-testing corrected p-value (FDR)
qvalue	FDR-related measure
geneID	DEGs contributing to the term
Count	Number of DEGs in term

13 Interpreting GeneRatio

Suppose:
- DEGs = 500
- Genes in DNA repair = 25

Then:

$$\text{GeneRatio} = \frac{25}{500} = 0.05$$

So:

5% of the DEG list is associated with that GO term.

However, **GeneRatio** alone does not establish significance. It must be interpreted with the background ratio and statistical test.

14 GO Results Example

GO Term	Category	Count	p.adjust
DNA repair	BP	35	0.001
Cell cycle	BP	42	0.003
DNA binding	MF	28	0.008
Nucleus	CC	75	0.0005
...	...	...	...

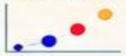
Interpretation:
DEGs are significantly associated with:
- DNA repair
- Cell cycle
- DNA binding
- Nuclear functions

15 GO Visualization

Common visualizations include:

Bar plot
Shows the most enriched GO terms.

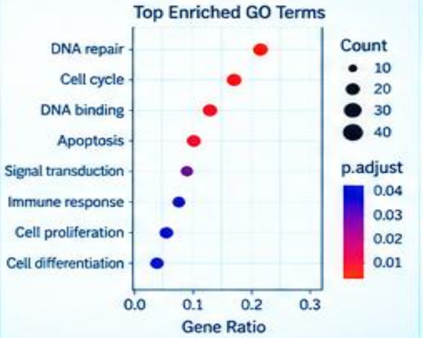
Dot plot
Displays enrichment, gene count and satistical significance.



Network plot
Shows relationships between GO terms and genes.

Enrichment map
Groups related GO terms.

16 GO Dot Plot



17 Upregulated vs Downregulated GO

Do not always analyze all DEGs together.
Perform separate analyses:

Upregulated DEGs → GO analysis → Enriched terms

Downregulated DEGs → GO analysis → Enriched terms

This can reveal different biological processes activated or suppressed.

18 GO Biological Interpretation

Upregulated DEGs
- Cell cycle
- DNA replication
- DNA repair

↓
Increased proliferative activity

Downregulated DEGs
- Cell differentiation
- Cell adhesion
- Normal tissue organization

↓
Loss of normal cellular functions

The biological interpretation should always be based on the actual genes and experimental context.

From Genes to Meaning!

1 KEGG: Basic Concept

What is KEGG?

Kyoto Encyclopedia of Genes and Genomes

KEGG is a database/resource for understanding biological systems through:

- Genes
- Proteins
- Metabolic pathways
- Signaling pathways
- Cellular processes
- Diseases
- Organism-specific pathways

Main question

"Which biological pathways are represented or enriched among my DEGs?"

2 GO vs KEGG

Gene Ontology (GO)	KEGG
Functional ontology	Pathway-oriented resource
BP / MF / CC	Signaling / metabolic pathways
Describes gene functions	Connects genes into pathways
Functional categories	Biological system-level interpretation
Example: DNA binding	Example: MAPK signaling

Simple distinction

GO = What does the gene do?

KEGG = Which pathway/system does the gene participate in?

GO Gene functions ⟷ KEGG Biological pathways

Complementary, not competitive!

3 KEGG Organization

KEGG contains several types of information.

KEGG Orthology (KO)

Groups genes with similar functions across organisms.

Pathways

Examples:

- MAPK signaling
- PI3K-Akt signaling
- Cell cycle
- Apoptosis
- Metabolic pathways

Disease pathways

Can connect molecular changes with disease-associated mechanisms.

KEGG links genes to pathways, pathways to diseases, and molecular data to biology.

4 KEGG Pathway Structure

Example: MAPK signaling pathway (simplified)

Growth factor → EGFR → RAS → RAF → MEK → ERK → Cell proliferation

- A pathway provides relationships among genes/proteins, rather than merely listing their functions.
- Pathways show how molecular events are connected.
- KEGG pathway maps can help interpret the biological context of DEGs.
- Many pathways include signaling, metabolic and regulatory components.

5 KEGG Analysis Input

Typical input:

Significant DEGs

Example:

TP53
EGFR
AKT1
PIK3CA
KRAS
MYC
...

Required information:

- Gene IDs
- Organism
- Background/universe gene set

Examples of IDs:

- Ensembl
- Entrez
- Gene Symbol
- UniProt

The genes need to be converted to an identifier recognized by the selected KEGG organism/database. For human analysis, the organism-specific KEGG identifier is:

hsa

6 KEGG Enrichment Workflow

DEG list → Filter significant → Gene IDs → ID conversion → KEGG mapping → Identify pathways → Enrichment analysis → Multiple-testing correction → Significant pathways → Visualization

7 KEGG Enrichment: Basic Principle

Suppose:

- Total background genes = 15,000
- DEGs = 500

A KEGG pathway contains:

- 100 genes

Among your DEGs:

- 30 genes

The question is:

- Is 30 genes an unusually large representation of this pathway?

If yes:

The pathway is significantly enriched.

8 KEGG Statistical Analysis

KEGG over-representation analysis can use the same general statistical framework as GO enrichment.

Hypergeometric / Fisher's exact test

Raw p-value → Multiple-testing correction → Adjusted p-value / FDR

A commonly used threshold is:

FDR < 0.05

The precise threshold should be defined before interpretation and reported with the analysis.

38 Final GO + KEGG Workflow

From Differentially Expressed Genes to Biological Mechanism

DEG DATA
(e.g., DESeq2 results)

Significant DEGs
(padj < 0.05, |log2FC| ≥ 1)

UP-regulated DEGs
(log2FC > 1)

DOWN-regulated DEGs
(log2FC < -1)

GO analysis
(BP / MF / CC)
Enriched GO terms for up-regulated genes

GO analysis
(BP / MF / CC)
Enriched GO terms for down-regulated genes

KEGG analysis
Enriched pathways for up-regulated genes

KEGG analysis
Enriched pathways for down-regulated genes

Compare pathways
- Common pathways
- Unique pathways
- Opposite regulation

Integrated analysis
- GO + KEGG integration
- Network analysis
- Key genes and pathways

Biological mechanism
- Functional interpretation
- Pathway-level understanding
- Hypothesis generation

From gene lists to biological insights!

39 Key Difference: GO vs KEGG

Gene Ontology (GO)

Focuses on Function

Gene Ontology
Unifying Biology

What does the gene do?
- What is the biological process?
- What is the molecular activity?
- Where does it function?

Three Ontologies

Biological Process (BP)
Biological processes or pathways in which a gene participates.

Molecular Function (MF)
Molecular activity of a gene product.

Cellular Component (CC)
Where the gene product is located or functions.

Examples
- Cell cycle
- DNA repair
- Apoptosis
- Signal transduction
- Nucleus
- Plasma membrane

KEGG

Focuses on Pathway/System

Kyoto Encyclopedia of Genes and Genomes

Which pathway does the gene participate in?
- Which signaling pathway?
- Which metabolic pathway?
- Which cellular pathway?
- How are genes connected?

Main components

KEGG Orthology (KO)
Groups genes with similar functions across organisms.

Pathways
Signaling, metabolic and cellular pathways.

Disease pathways
Links molecular changes with diseases.

Examples
- MAPK signaling pathway
- PI3K-Akt signaling pathway
- Cell cycle
- Apoptosis
- Metabolic pathways
- Cancer pathways

Together:
GO explains functional categories;
KEGG provides pathway-level context.

Complementary, not competitive!

40 Final Take-Home Message

From DEG to Biological Mechanism

DESeq2	• Differential expression analysis • Identify significant genes
DEGs	• Up-regulated and down-regulated genes • padj < 0.05, |log2FC| ≥ 1
UniProt / BLAST	• Obtain protein sequences • Identify UniProt accession
Protein annotation	• Protein function, domains • Subcellular location, cross-references
GO analysis	• Functional enrichment (BP/MF/CC) • Identify enriched GO terms
KEGG analysis	• Pathway enrichment • Identify significant pathways
GO + KEGG integration	• Compare and integrate results • Common and unique pathways • Network analysis
Biological mechanism	• Functional and pathway interpretation • Key genes and biological insights • Hypothesis generation

Core principle

Do not stop at "which genes are differentially expressed."

The goal of downstream annotation is to determine:

"What functions and pathways are altered, which genes drive those changes, and how they may explain the biological phenotype?"

From gene lists to mechanisms, from data to discovery!

Mail id- manabendranath5017@gmail.com
Contact No- 8638068114/7399423215